

Motion of particles on a friction surface

Newton's apple Team 13

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Introduction & Contextual Background

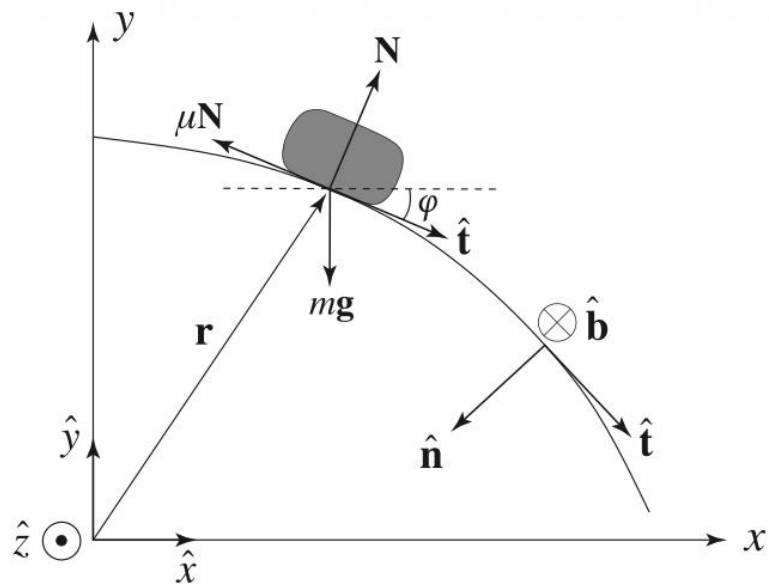
Introduction

The motion of an object sliding on a curved surface is a classical mechanics problem involving gravity, friction, and surface geometry. This study is based on *Motion of a Block Slipping on an Arbitrary Surface with Friction* (Gonzalez-Cataldo et al., 2017), which presents a general framework for predicting when an object loses contact with a curved surface.

The model relates the particle's motion to the local curvature of the surface and introduces a horizon function to determine the departure point.

In this project, the theory was applied to a hemispherical acrylic surface and compared with experimental measurements.

Kinematics



$$\mathbf{r}(t) = (x(t), y(t))$$

$$s(t) = \int_{t_0}^t \left\| \frac{d\mathbf{r}(\tau)}{d\tau} \right\| d\tau, \quad \text{where } t \in [t_0, T],$$

$$\hat{\mathbf{t}} \equiv \frac{d\mathbf{r}}{ds},$$

$$\hat{\mathbf{n}} \equiv \frac{1}{\kappa(s)} \frac{d\hat{\mathbf{t}}}{ds},$$

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$$\hat{\mathbf{b}} \equiv \hat{\mathbf{t}} \times \hat{\mathbf{n}},$$

Kinematics

$$\mathbf{v}(s) = v(s)\hat{\mathbf{t}}(s)$$

$$\mathbf{a}(s) = v(s)\frac{dv(s)}{ds}\hat{\mathbf{t}}(s) + v^2(s)\kappa(s)\hat{\mathbf{n}}(s).$$

- Velocity is always directed along the tangent vector.
- The first term represents the tangential acceleration (change in speed).
- The second term represents the normal acceleration (change in direction).

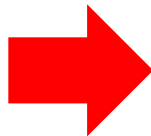
Dynamics in the Frenet-Serret frame

⟨the tangential equation of motion⟩

$$mv(s) \frac{dv(s)}{ds} = m\mathbf{g} \cdot \hat{\mathbf{t}}(s) - \mu N(s)$$

⟨the normal-direction equation of motion⟩

$$m\kappa(s)v^2(s) = m\mathbf{g} \cdot \hat{\mathbf{n}}(s) + (\hat{\mathbf{b}} \cdot \hat{\mathbf{z}})N(s).$$



⟨the most general equation representing the velocity of a particle moving on a curved surface⟩

$$v^2(s) = v_0^2 e^{-(2\mu/\hat{\mathbf{b}} \cdot \hat{\mathbf{z}})\Phi(s)} + 2 e^{-(2\mu/\hat{\mathbf{b}} \cdot \hat{\mathbf{z}})\Phi(s)} \times \int_0^s \left[\mathbf{g} \cdot \hat{\mathbf{t}}(s') + \mu \frac{\mathbf{g} \cdot \hat{\mathbf{n}}(s')}{\hat{\mathbf{b}} \cdot \hat{\mathbf{z}}} \right] e^{(2\mu/\hat{\mathbf{b}} \cdot \hat{\mathbf{z}})\Phi(s')} ds'$$

Tangent angle

Eq.01

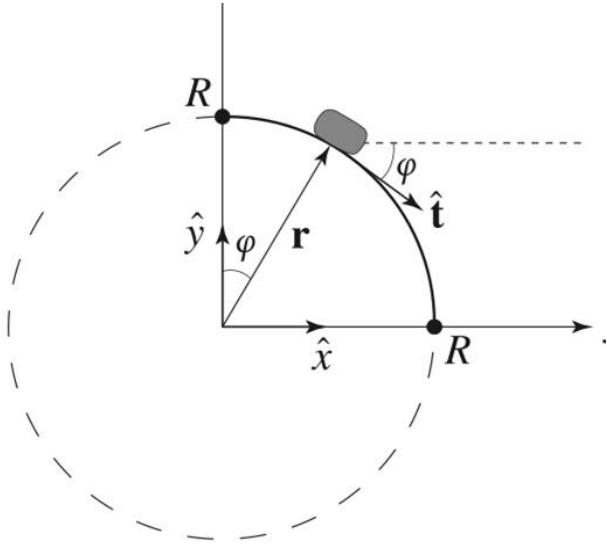
$$v^2(\varphi) = v_0^2 e^{2\mu(\varphi-\varphi_0)} + 2ge^{2\mu\varphi} \times \int_{\varphi_0}^{\varphi} (\sin \varphi' - \mu \cos \varphi') \frac{e^{-2\mu\varphi'}}{\kappa(\varphi')} d\varphi';$$

Eq.02

$$H(\varphi) = \frac{1}{gR} \frac{\mathbf{g} \cdot \hat{\mathbf{n}}}{\kappa(\varphi)} = \frac{\cos \varphi}{R\kappa(\varphi)}$$

- By introducing the tangent angle, the particle velocity can be expressed as a function of the local geometry of the curve.
- Eq.01 describes how the velocity evolves along the surface.
- Eq.02 defines the horizon function , which determines the condition for losing contact with the surface.
- These equations simplify the analysis of motion on curved surfaces.

Departure from the circle



$\mathbf{r}(\varphi) = (R \sin \varphi, R \cos \varphi)$, where $\varphi \in [0, \pi/2]$.

$$\kappa(\varphi) = 1/R$$

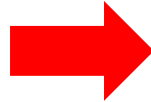


$$H(\varphi) = \cos \varphi$$

Departure from the circle

⟨Condition for Leaving the Surface⟩

$$\frac{v^2(\varphi)}{gR} \leq \cos \varphi,$$



$$\cos \varphi = \frac{2}{3} + \frac{v_0^2}{3gR},$$

⟨Special Case⟩ $\mu = 0,$

$$v^2(\varphi) = v_0^2 + 2gR(1 - \cos \varphi),$$

Experimental process

Material lists

- Rectangular parallelepiped blocks Transparent hemisphere (21 cm in diameter)
- Digital camera
- Inclined plane (constructed from an L-shaped plastic folder)
- Measuring tape

Experimental procedure

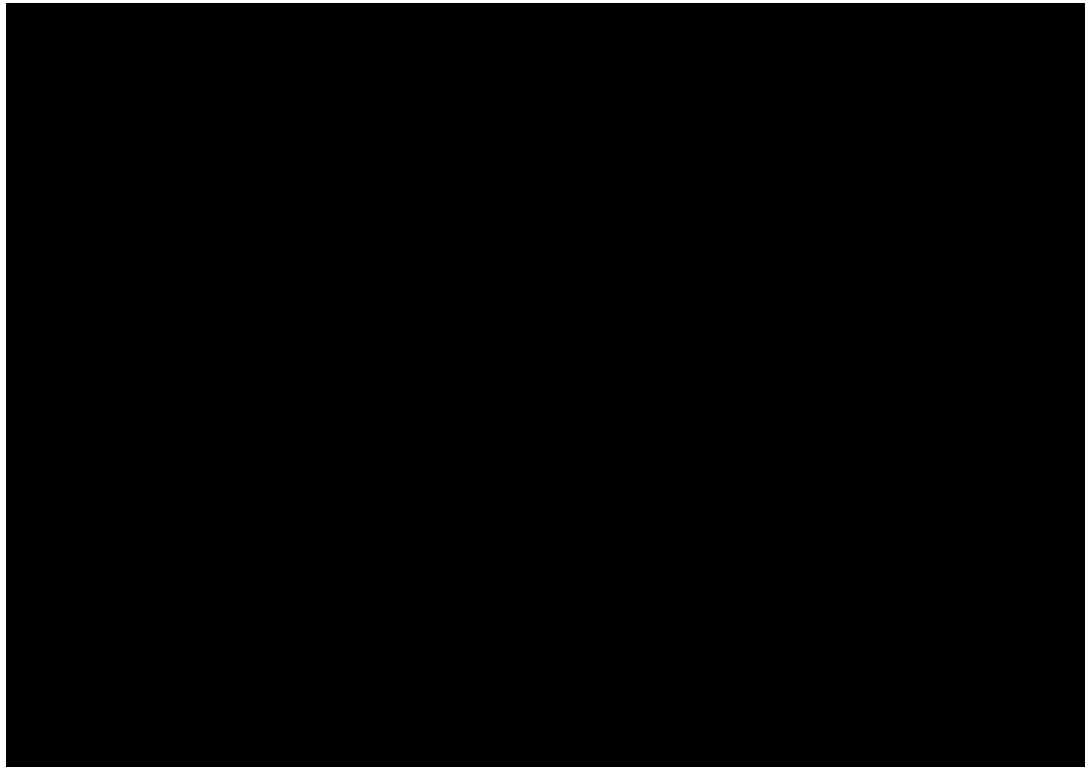
- 1. The initial drop height of the object is measured.**
- 2. The inclined plane is secured to the apex of the transparent hemisphere.**
- 3. A camera is positioned perpendicularly to the ground to record the experiment.**
- 4. The block is released from the inclined plane to slide downward.**
- 5. The initial velocity at the apex of the hemisphere is calculated using the law of conservation of mechanical energy.**
- 6. The detachment angle, at which the block separates from the surface of the hemisphere, is measured through video analysis.**

Experimental procedure

- 7. The velocity at the exact moment of detachment is determined using the position and time coordinates of the block at an arbitrary point after separation.**
- 8. The friction coefficient of the acrylic hemisphere is calculated by substituting the obtained values into the governing equations from previous literature.**
- 9. The experimental error is determined by comparing the measured friction coefficient of the acrylic hemisphere with its theoretical value.**
- 10. The potential sources of experimental error are analyzed and discussed**

Experimental result

Our Experiment Video



Determining the Departure Point of the Block

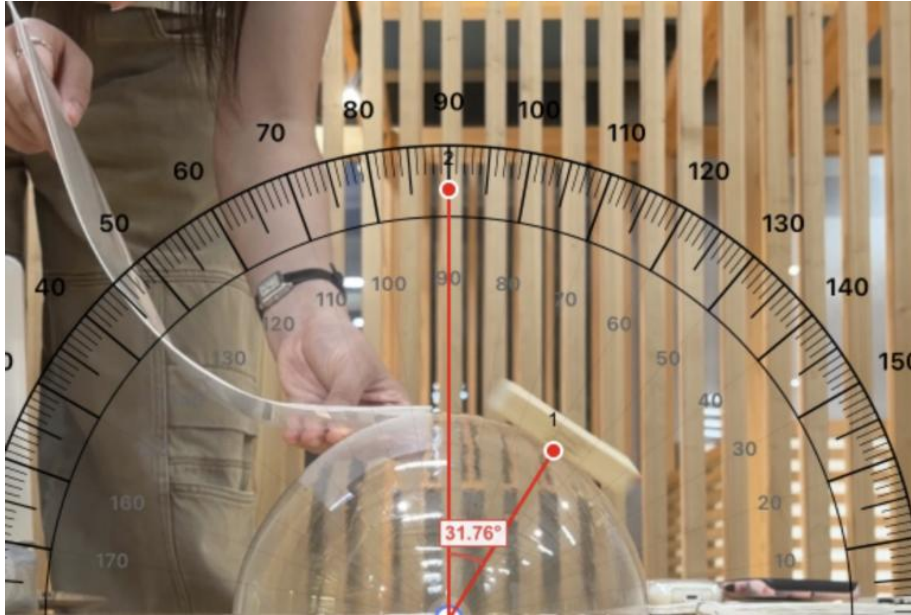


figure 01: Point A

〈Measured Angle〉

$$\varphi = 31.76^\circ = 0.5543 \text{ rad}$$

〈Height at Point A〉

$$\begin{aligned} x(A) &= R \cos \varphi \\ &= 0.101 \times \cos(31.76^\circ) = 0.08588 \text{ m} \end{aligned}$$

〈Horizontal Distance〉

$$\begin{aligned} y(A) &= R \sin \varphi \\ &= 0.101 \times \sin(31.76^\circ) = 0.05316 \text{ m} \end{aligned}$$

〈Measured Time〉

$$t(A) = 0.42 \text{ s}$$

Calculating the Velocity at the Departure point

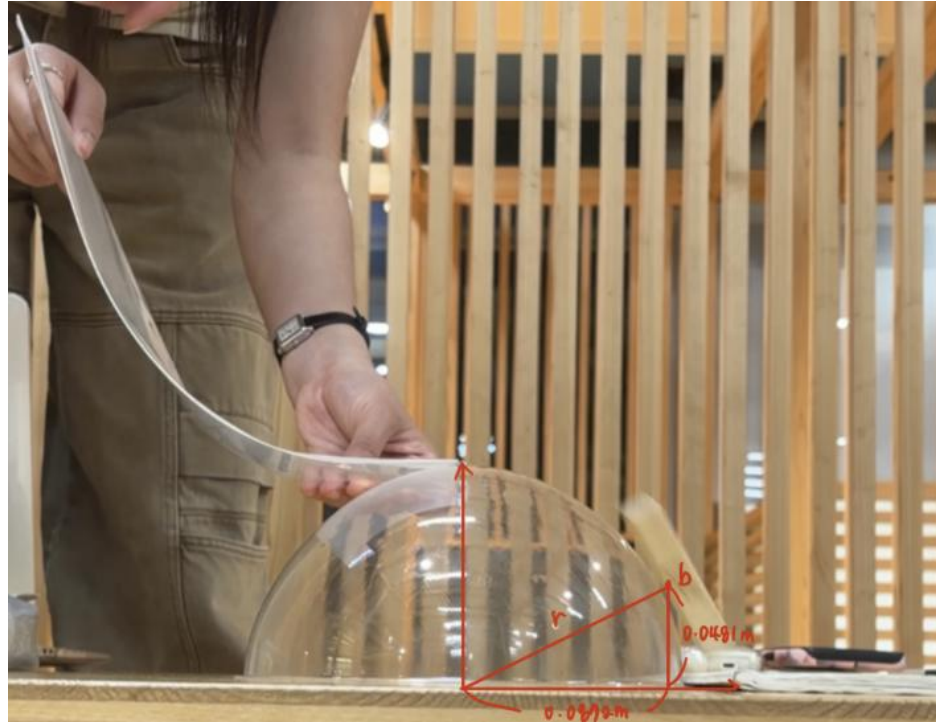


figure 02: Point B

Calculating the Velocity at the Departure point

⟨Time Difference⟩

$$\Delta t = 0.47 - 0.42 = 0.05 \text{ s}$$

⟨Horizontal Displacement⟩

$$\begin{aligned}\Delta x &= 0.08722 - 0.05316 \\ &= 0.03406 \text{ m}\end{aligned}$$

⟨Vertical Displacement⟩

$$\begin{aligned}\Delta y &= 0.08588 - 0.04810 \\ &= 0.03778 \text{ m}\end{aligned}$$

⟨Horizontal Velocity⟩

$$v_x = \Delta x / \Delta t = 0.03406 / 0.05 = 0.6812 \text{ m/s}$$

⟨Vertical Velocity⟩

$$\Delta y = v_y \Delta t + \frac{1}{2} g \Delta t^2$$

$$0.03778 = v_y(0.05) + \frac{1}{2}(9.8)(0.05)^2$$

$$v_y = 0.5106 \text{ m/s}$$

⟨Speed at Point A⟩

$$\begin{aligned}v_A &= \sqrt{v_x^2 + v_y^2} = \sqrt{[(0.6812)^2 + (0.5106)^2]} \\ &= 0.8513 \text{ m/s}\end{aligned}$$

Calculating the initial Velocity Using Energy Conservation

⟨Given⟩

- Mass

$$m = 0.0173 \text{ kg}$$

- Initial Height

$$h_0 = 0.288 \text{ m}$$

- Hemisphere Height

$$h_{\text{top}} = 0.101 \text{ m}$$

⟨Initial Potential Energy⟩

$$\begin{aligned} PE_{\text{initial}} &= mgh \\ &= (0.0173)(9.8)(0.288) \\ &= 0.04883 \text{ J} \end{aligned}$$

⟨Potential Energy at Top⟩

$$\begin{aligned} PE_{\text{top}} &= (0.0173)(9.8)(0.101) \\ &= 0.01712 \text{ J} \end{aligned}$$

⟨Kinetic Energy⟩

$$\begin{aligned} KE &= 0.04883 - 0.01712 = \\ &0.03170 \text{ J} \end{aligned}$$

Initial Velocity

$$\begin{aligned} KE &= \frac{1}{2}mv^2 \\ v_0 &= \sqrt{(2KE/m)} = 1.914 \text{ m/s} \end{aligned}$$

Applying the AJP Analytical Model

$$\frac{v^2(\varphi)}{gR} = \frac{v_0^2}{gR} e^{2\mu\varphi} + \frac{(2 - 4\mu^2)(e^{2\mu\varphi} - \cos \varphi) - 6\mu \sin \varphi}{1 + 4\mu^2}$$

⟨Known Values⟩

$$v(\varphi) = 0.8513 \text{ m/s}$$

$$v_0 = 1.914 \text{ m/s}$$

$$\varphi = 0.5543 \text{ rad}$$

$$R = 0.101 \text{ m}$$

$$g = 9.8 \text{ m/s}^2$$

figure 03: Equation from the Paper

Want to know : μ

Solving for the Friction Coefficient

⟨Numerical Solution⟩

Substituting all known values into the model : $\mu = -1.17$

⟨Comparison⟩

Expected Acrylic Friction : $\mu \approx 0.83$

Experimental Result : $\mu = -1.17$

Conclusion

Why Did We Obtain a Negative Friction Coefficient ?

- **Source 1 : Overestimated Initial Velocity**

Assumption : No energy loss before reaching the hemisphere

Reality : Friction existed on the guiding track

→ **v_0 too large : 1.914 m/s**

- **Source 2 : Underestimated Departure Velocity**

Sources of error : Air resistance, Pixel scaling, Motion blur, Perspective distortion

→ **v_A too small : 0.851 m/s**

Why Did We Obtain a Negative Friction Coefficient ?

⟨Result⟩

Overestimated v_0 + Underestimated v_A



Negative friction coefficient: $\mu = -1.17$

Conclusions

⟨Achievements⟩

- ✓ Identified the departure point experimentally
- ✓ Calculated departure velocity using projectile motion
- ✓ Calculated initial velocity using energy conservation
- ✓ Applied the AJP analytical model
- ✓ Investigated the discrepancy between theory and experiment

Conclusions

⟨Final Takeaway⟩

The most important outcome was not the numerical value of μ but, understanding how the intermediate mathematical operations connect theoretical equations to real physical motion.

Thank you.